



**PUNJAB PUBLIC SERVICE COMMISSION**  
**COMBINED COMPETITIVE EXAMINATION**  
**FOR RECRUITMENT TO THE POSTS OF**  
**PROVINCIAL MANAGEMENT SERVICE, ETC -2021**  
**CASE NO. 3C2022**

**SUBJECT: ZOOLOGY (PAPER-I)**

**TIME ALLOWED: THREE HOURS**

**MAXIMUM MARKS: 100**

**NOTE:**

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

**NOTE:** Attempt FIVE Questions in all. Select at least TWO Questions from each Section. Attempt in Urdu or English.

**SECTION-I**

- Q.No.1.** (a) What are the different modes of locomotion in Protozoa?  
(b) Discuss in detail the shell in different groups of Molluscs.  
(10+10=20 Marks)
- Q.No.2.** (a) Describe the general characteristics of phylum Aschelminthes.  
(b) Discuss different types of reproduction found in Porifera.  
(10+10=20 Marks)
- Q.No.3.** (a) Explain the water vascular system in Echinodermata.  
(b) Write note on larva of various classes of Arthropoda.  
(10+10=20 Marks)
- Q.No.4.** Write short note on any FOUR of the following: (4 x 5 = 20 Marks)  
a. Tagmatization      b. Metamorphosis      c. Nematocysts  
d. Polymorphism      e. Body plan of triploblastic acoelomate

**SECTION-II**

- Q.No.5.** (a) Write a detailed note on earliest known vertebrate.  
(b) Discuss the adaptations in external structure and locomotion in mammals.  
(10+10=20 Marks)
- Q.No.6.** (a) What adaptations of birds promote endothermy and flight? Why is endothermy Important for birds?  
(b) Explain in detail the causes of evolution and extinction of dinosaurs.  
(10+10=20 Marks)
- Q.No.7.** (a) Discuss the origin and evolution of mammals tracing their synapsid lineage.  
(b) Describe the process of excretion and osmoregulation in amphibians.  
(10+10=20 Marks)
- Q.No.8.** Write short note on any FOUR of the following: (4x5=20 Marks)  
a. Feathers      b. Migration of birds      c. Endothermy  
d. Specialized teeth in mammals      e. Evolution of jaw in vertebrates

  
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**CASE NO. 3C2022**

**SUBJECT: ZOOLOGY (PAPER-II)**

**TIME ALLOWED: THREE HOURS**

**MAXIMUM MARKS: 100**

**NOTE:**

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

**NOTE:** **Attempt FIVE Questions in All Selecting ONE from each Section. Attempt in Urdu or English.**

**SECTION-A**

- Q. No. 1.** a) Differentiate between reversible and irreversible enzyme inhibition.  
b) Differentiate between catabolic and anabolic pathways of metabolism.  
(10 + 10 = 20 Marks)

- Q. No. 2.** a) Explain the series of chemical reactions occurring during Krebs cycle.  
b) What is meiotic nondisjunction? What are its consequences for human beings?  
(10 + 10 = 20 Marks)

**SECTION-B**

- Q. No. 3.** a) How has excretion of nitrogenous wastes evolved in aquatic and terrestrial animals?  
b) Which different mechanisms animals use to regulate the exchange of heat between themselves and the environment?  
(10 + 10 = 20 Marks)

- Q. No. 4.** a) Describe the motility of gastrointestinal tract in animals.  
b) Name neurohormones released from the mammalian posterior pituitary gland (neurohypophysis), their target sites and functions.  
(10 + 10 = 20 Marks)

**SECTION-C**

- Q. No. 5.** a) Define multiple alleles. Explain a case example of multiple alleles.  
b) Give a brief account of the recessive genetic disorders (recessively inherited disorders).  
(10 + 10 = 20 Marks)

- Q. No. 6.** a) What is a genetic code? Which general features characterize genetic code?  
b) What are the transgenic animals? How are they produced and what are their uses?  
(10 + 10 = 20 Marks)

**SECTION-D**

- Q. No. 7.** a) How has Neo-Darwinism validated the Darwinism? Explain Darwin's theory in the light of Neo-Darwinism.  
b) How natural selection leads to the formation of adaptations? Explain with an example.  
(10 + 10 = 20 Marks)

- Q. No. 8.** a) Differentiate between natural selection and genetic drift.  
b) What is Hardy-Weinberg equilibrium? Explain different factors which act as sources of microevolutionary change.  
(10 + 10 = 20 Marks)

**SECTION-E**

- Q. No. 9.** a) Describe a closed ecological system.  
b) Discuss the efficiency of energy transfer in ecosystems.  
(10 + 10 = 20 Marks)

- Q. No. 10.** a) How is nutrient cycling connected with pollution? Explain relationship between carbon cycle and global warming.  
b) How do animals adapt themselves to their habitats? What are animals' adaptations to desert environment?  
(10 + 10 = 20 Marks)